1 Taylor expansion at order n (Taylor–Young formula)

RESULT

$$f(a+h) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}\, h^k + o(h^n) \qquad \text{as } h \to 0$$

Setting and notation. Let $I \subset \mathbb{R}$ be an open interval, $a \in I$ and $n \geq 1$ an integer. Let $f : I \to \mathbb{R}$ be n times differentiable at a, meaning that $f, f', \ldots, f^{(n-1)}$ exist on a neighbourhood of a and that $f^{(n-1)}$ is differentiable at a. We write $f^{(k)} = \frac{d^k f}{dx^k}$, with $f^{(0)} = f$, and we define, for h such that $a + h \in I$,

$$P_n(h) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}\, h^k \qquad \text{and} \qquad R_n(h) = f(a+h) - P_n(h).$$

Recall that $R_n(h) = o(h^n)$ means: for every $\varepsilon > 0$ there exists $\delta > 0$ such that $|h| < \delta \Rightarrow |R_n(h)| \leq \varepsilon\, |h|^n$. We must therefore prove that $R_n(h) = o(h^n)$. We proceed by induction on n.

Tool: the mean value theorem. If g is continuous on the closed interval between 0 and h and differentiable inside it, there exists c strictly between 0 and h such that

$$g(h) - g(0) = g'(c)\, h.$$

Step 1: initialisation ($n = 1$). Here $P_1(h) = f(a) + f'(a)\, h$. For $h \neq 0$,

$$\frac{R_1(h)}{h} = \frac{f(a+h) - f(a)}{h} - f'(a) \xrightarrow[h \to 0]{} f'(a) - f'(a) = 0,$$

by definition of the derivative $f'(a)$. Hence $R_1(h) = o(h)$: the property holds for $n = 1$.

Step 2: induction hypothesis. Fix $n \geq 1$ and assume that the result holds at order n for *every* function that is n times differentiable at a. Let f be $n+1$ times differentiable at a. We show that $R_{n+1}(h) = o(h^{n+1})$.

Step 3: apply the hypothesis to f'. Since f is $n+1$ times differentiable at a, its derivative f' is n times differentiable at a, with $(f')^{(k)} = f^{(k+1)}$. The induction hypothesis applied to f' gives

$$f'(a+h) = \sum_{k=0}^{n} \frac{f^{(k+1)}(a)}{k!}\, h^k + o(h^n). \tag{$*$}$$

Step 4: differentiate the remainder. The remainder R_{n+1} is differentiable on a neighbourhood of 0, since f' exists on a neighbourhood of a. Differentiating term by term, with $\frac{d}{dh}\frac{h^k}{k!} = \frac{h^{k-1}}{(k-1)!}$ for $k \geq 1$ and the constant term vanishing:

$$\begin{aligned} R'_{n+1}(h) &= f'(a+h) - \sum_{k=1}^{n+1} \frac{f^{(k)}(a)}{(k-1)!}\, h^{k-1} \\ &= f'(a+h) - \sum_{j=0}^{n} \frac{f^{(j+1)}(a)}{j!}\, h^j \qquad \text{(change of index } j = k-1) \\ &= o(h^n) \qquad \text{by } (*). \end{aligned}$$

Moreover $R_{n+1}(0) = f(a) - f^{(0)}(a) = 0$.

Step 5: go back to R_{n+1} with the mean value theorem. Let $\varepsilon > 0$. By Step 4, there exists $\delta > 0$, small enough for R_{n+1} to be differentiable on $]-\delta, \delta[$, such that

$$|u| < \delta \implies |R'_{n+1}(u)| \leq \varepsilon\, |u|^n.$$

Take h with $0 < |h| < \delta$. Applying the mean value theorem to R_{n+1} between 0 and h, there exists c_h strictly between 0 and h such that

$$R_{n+1}(h) = R_{n+1}(h) - R_{n+1}(0) = R'_{n+1}(c_h)\, h.$$

Since $|c_h| < |h| < \delta$,

$$|R_{n+1}(h)| = |R'_{n+1}(c_h)|\,|h| \;\leq\; \varepsilon\,|c_h|^n\,|h| \;\leq\; \varepsilon\,|h|^n\,|h| = \varepsilon\,|h|^{n+1}.$$

This holds for every $\varepsilon > 0$, hence $R_{n+1}(h) = o(h^{n+1})$: the result holds at order $n+1$.

Step 6: conclusion. The property is true for $n = 1$ and passes from n to $n+1$. By induction, for every $n \geq 1$ and every function f that is n times differentiable at a,

$$f(a+h) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}\,h^k + o(h^n) \qquad \text{as } h \to 0. \qquad \blacksquare$$

2 Continuous compounding via a differential equation

RESULT

$$M_t = M_0\,e^{rt}$$

Proof. Let $M : \mathbb{R}_+ \to \mathbb{R}_+$, $t \mapsto M(t)$, be a function of class $\mathcal{C}^1$ on $\mathbb{R}_+$, representing the amount held by the investor at time t. Applying a first-order Taylor expansion, for all $t \in \mathbb{R}_+$ and as $dt \to 0$:

$$M(t+dt) = M(t) + \frac{dM}{dt}(t)\,dt + o(dt) \quad \Longleftrightarrow \quad \underbrace{M(t+dt) - M(t)}_{=\Delta M} = \frac{dM}{dt}(t)\,dt + o(dt).$$

On the other hand, the change in M over dt is the interest earned during this period, which is proportional to r, $M(t)$ and dt. Hence,

$$\Delta M = r\,M(t)\,dt.$$

Finally, we obtain

$$\begin{aligned} & \frac{dM}{dt}(t)\,dt + o(dt) = r\,M(t)\,dt \\ \Longleftrightarrow \quad & \frac{dM}{dt}(t) + \frac{o(dt)}{dt} = r\,M(t) \\ \underset{dt\to 0}{\Longleftrightarrow} \quad & \frac{dM}{dt}(t) = r\,M(t). \end{aligned}$$

This is a simple first-order linear ordinary differential equation, whose solution is

$$M_t = M_0\,e^{rt}. \qquad \blacksquare$$